



PowerPoint Presentation by
Gail B. Wright
Professor Emeritus of Accounting
Bryant University

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MANAGEMENT ACCOUNTING

8th EDITION

BY

HANSEN & MOWEN

3 ACTIVITY COST BEHAVIOR

LEARNING OBJECTIVES

After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Define cost behavior for fixed, variable, & mixed costs.
2. Explain the role of the resource usage model in understanding cost behavior.
3. Separate mixed costs into their fixed & variable components using the high-low method, scatterplot method, and method of least squares.

Continued

LEARNING OBJECTIVES

4. Evaluate the reliability of a cost equation.
5. Discuss the role of multiple regression in assessing cost behavior.
6. Describe the use of managerial judgment in determining cost behavior.

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT: Reddy Heaters

If the division reduces demand for rework activity, will resource spending be reduced by the same proportion? Is there a difference between resource spending and resource usage?

QUESTIONS TO THINK ABOUT: Reddy Heaters

If total cost of rework and # of units reworked are known, is it possible to determine how much is variable cost? How much is fixed cost? Is knowing variable- and fixed-cost behavior important?

QUESTIONS TO THINK ABOUT: Reddy Heaters

What role does management play in determining cost behavior?

QUESTIONS TO THINK ABOUT: Reddy Heaters

Can you think of reasons other than those in the scenario that make it important for managers to understand cost behavior?



LEARNING OBJECTIVE

1

Define cost behavior for fixed, variable, & mixed costs.

If Reddy Heaters produces twice as many heaters as last year, will production costs double?

NO. *Variable* costs will double if production doubles but **fixed** costs will not change.



REDDY HEATERS:

Cutting Activity

- ✧ Cutting activity uses 2 inputs
 - ✧ Cutting machine
 - ✧ 1 machine can produce up to 240,000, 3-inch segments per year (fixed cost)
 - ✧ Power to operate machine (variable cost)



FIXED COST: Definition

Fixed costs do not vary over the relevant range.

Reddy Heaters: 1 cutting machine costs \$60,000 per year & can produce up to 240,000, 3-inch segments



VARIABLE COST: Definition

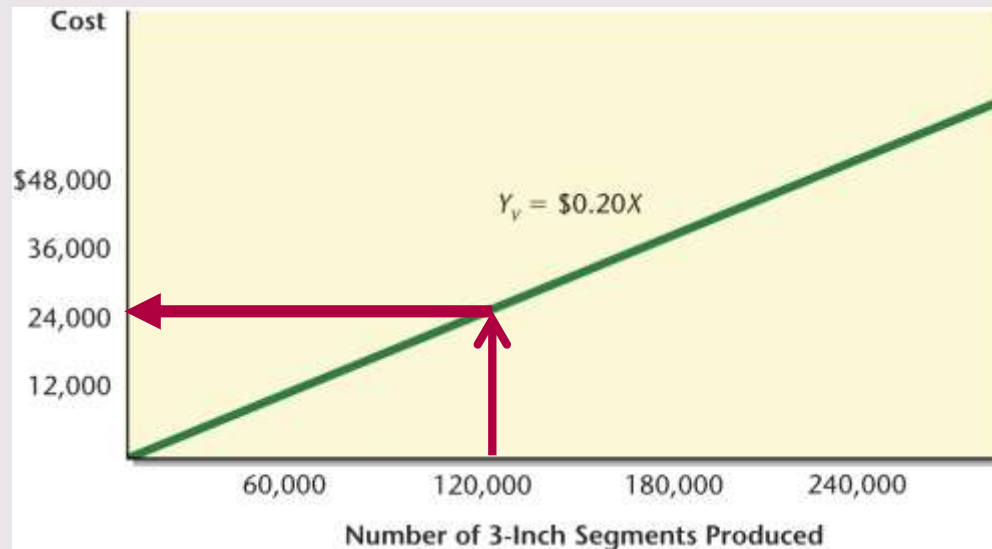
Variable costs vary in direct proportion to changes in output.

Reddy Heaters: 1 segment uses 0.1 kilowatts at cost of \$2.00 per kilowatt. Each segment costs \$0.20.



VARIABLE COSTS:

\$0.20 per Unit Produced



What is the total variable cost to produce 120,000 3-inch segments?

EXHIBIT 3-2

MIXED COST: Definition

Mixed costs have a variable and a fixed component.

Reddy Heaters: sales people earn a \$10,000 salary + \$0.50 commission on each heater sold.



COST BEHAVIOR ACTIVITIES

- ✧ Every activity has a
 - ✧ Time horizon for measurement
 - ✧ Resources to accomplish the task
 - ✧ Materials
 - ✧ Labor
 - ✧ Capital
 - ✧ Output measures (activity drivers)



ACTIVITY DRIVERS 1

- ✧ Production drivers (unit level)
 - ✧ Explain changes in unit cost as units produced changes
 - ✧ Are inputs with direct relationship with production level
 - ✧ Examples:
 - ✧ Pounds direct materials
 - ✧ Direct labor



ACTIVITY DRIVERS 2

- ✧ Non-unit level activity drivers
 - ✧ Explain changes in cost in terms other than changes in units of production
 - ✧ Have no direct relationship with production
 - ✧ Examples
 - ✧ Depreciation
 - ✧ Set-up costs incurred to change the items produced



ACTIVITY DRIVERS & COST BEHAVIOR

* FBM

- * Functional-based cost system includes only unit-level costs in observations of cost behavior

* ABM

- * Activity-based cost system includes both unit- and non-unit level costs in observations of cost behavior

* Implication

- * ABM produces richer view of cost behavior



LEARNING OBJECTIVE

2

Explain the role of resource usage model in understanding cost behavior.



CAPACITY: Definition

Capacity for an activity is the amount of an activity a company can perform.

Practical capacity is the level at which company can perform *efficiently*.



How much capacity does a company need? What happens if there is excess capacity?

Need for capacity depends on level of performance required.
Excess capacity affects cost behavior.



FLEXIBLE RESOURCES

- ✧ Are resources that can be acquired **as needed**
- ✧ No long term commitment
- ✧ Quantity supplied = quantity demanded
 - ✧ >>>>> NO EXCESS CAPACITY
- ✧ Example: direct materials



COMMITTED RESOURCES

- ✧ Are resources acquired **in advance** of usage
- ✧ Often have long term commitment
- ✧ Quantity supplied (often) $\neq$ quantity demanded
 - ✧ >>>>> MAY MEAN EXCESS CAPACITY
- ✧ Example: factory building



COMMITTED RESOURCES: Can Be

Committed fixed costs, such as a building or equipment bought, leased; or

Committed discretionary costs, such as implicit contracts with employees.



STEP COST: Definition

Step-costs exhibit a discontinuous behavior pattern.

Step-costs are constant for a certain range of output, then jump to another level, remaining constant again over a certain range of output.



CHANGE ORDER EQUATIONS

$$\text{CHANGE ORDER} = \text{Fixed Cost} + \text{Variable Cost}$$

$$= \text{Engineering Cost} + \text{Supply Cost}$$

$$\text{Fixed activity rate} = \frac{\text{Total committed cost}}{\text{Total capacity available}}$$

$$\text{Variable activity rate} = \frac{\text{Total cost of flexible resources}}{\text{Capacity used}}$$



IMPLICATIONS

- ✧ Improving managerial control & decision making by
 - ✧ Encouraging managers to pay more attention to controlling resource usage, spending
 - ✧ Providing information to control capacity efficiently
 - ✧ Allowing managers to calculate change in resource supply, demand



LEARNING OBJECTIVE

3

Separate mixed costs into fixed & variable components using high-low, scatterplot, & least squares.



LINEARITY ASSUMPTION

Variable cost assumes a **linear relationship** between cost and activity driver.

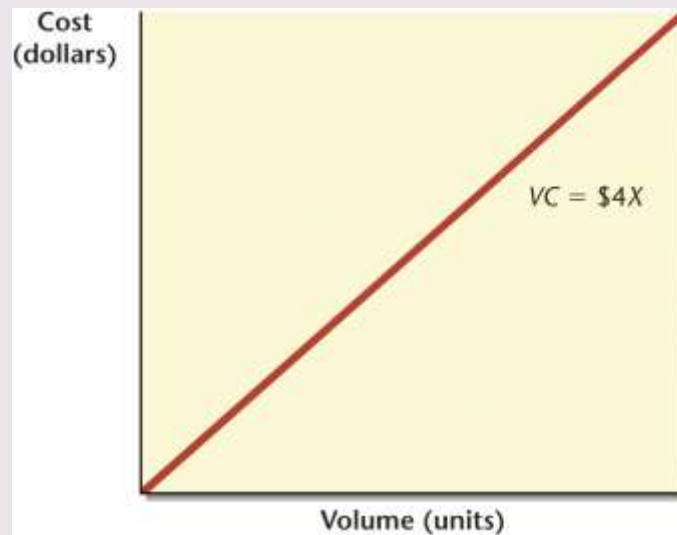


EXHIBIT 3-7

$$\text{TOTAL COSTS} = \text{Fixed cost} + (\text{Variable rate} \times \text{Output})$$

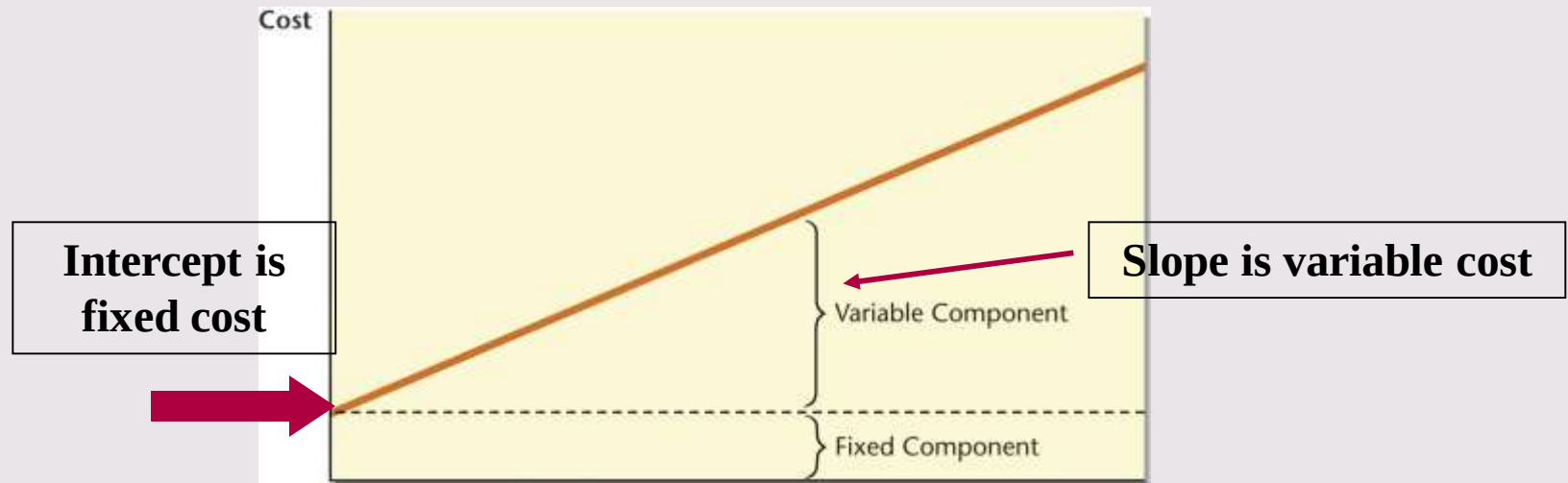


EXHIBIT 3-10

HIGH-LOW EQUATIONS

Variable rate = Change in cost / Change in output
(High cost – Low cost) / (High output – Low output)

Fixed cost =
Total cost for High (**Low**) point –
{ Variable rate x High (**Low**) output }

SCATTERPLOT METHOD

Scatterplot is a method of determining the equation of a line by plotting the data on a graph.

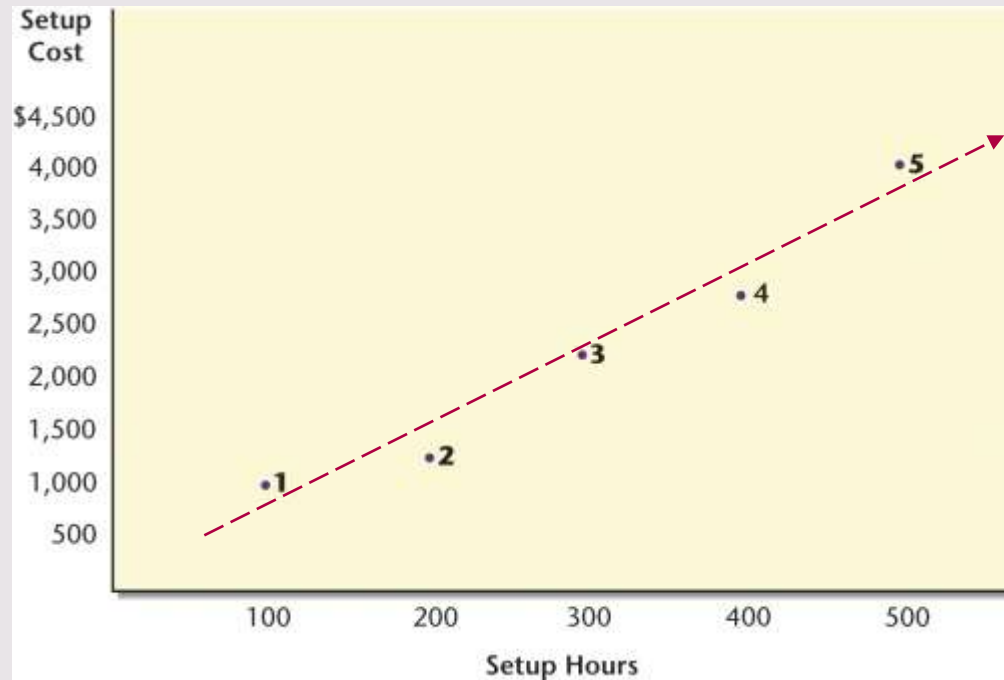


EXHIBIT 3-11

What are the advantages,
disadvantages of scatterplot?

Scatterplot

- ✧ Allows you to see the data BUT
- ✧ It lacks any objective criterion for choosing the best-fitting line

METHOD OF LEAST SQUARES

Squares the vertical deviations from point in scatterplot to line drawn by least squares methods, then adds them to give an overall measure of closeness of fit of line to data.



LEARNING OBJECTIVE

4

Evaluate the reliability of a cost equation.



How reliable is the cost equation developed by the least squares method?

R^2 , the coefficient of determination, and the **coefficient of correlation** will tell you the goodness of fit of your cost equation.



COEFFICIENT OF DETERMINATION (R^2)

- ✳ Percentage of variability in dependent variable explained by independent variable
 - ✳ Range: 0 – 1
 - ✳ Higher is better

COEFFICIENT OF CORRELATION

- ✧ Square root of coefficient of determination
- ✧ Measures whether variables move in same (+) or opposite (-) directions
 - ✧ Range: -1 - +1

LEARNING OBJECTIVE

5

Discuss the role of multiple regression in assessing cost behavior.



MULTIPLE REGRESSION: Definition

Multiple regression uses 2 or more independent variables (variable costs) in addition to the y-intercept (fixed cost) to explain the dependent variable.



LEARNING OBJECTIVE

6

Describe the use of managerial judgment in determining cost behavior.



MANAGERIAL JUDGMENT

- ✧ Is a method of cost assignment used to
 - ✧ Determine fixed, variable cost
 - ✧ Uses managerial experience
 - ✧ Uses past observation of cost relationships
 - ✧ To refine statistical estimation results
- ✧ Advantage: simplicity
- ✧ Disadvantage: judgment errors



CHAPTER 3

THE END